

Review

Review of the development of power system out-of-step splitting control and some thoughts on the impact of large-scale access of renewable energy

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ABSTRACT

Out-of-step oscillation is a very destructive physical phenomenon in power system, which could directly cause big blackout accompanied by serious sociology-economic impacts. Out-of-step splitting control is an indispensable means, which could protect the system from major shocks of out-of-step oscillation. After years of development, it has achieved certain amount of research results. Have the existing methods been able to meet the requirements of out-of-step splitting? What improvements are needed? Under this background, this review is written. It combs the development of out-of-step splitting control technologies and analyzes the technical routes and characteristics of different methods. It points out the contradiction between rapidity and optimality is the biggest technical problem, existing in both the traditional local measurement based out-of-step splitting protection and the wide-area information based out-of-step splitting protection. It further points out that the advantages of the two types of protections can be combined with the unique physical characteristics of the out-of-step center to form a more advantageous splitting strategy. Besides, facing the fact of large-scale renewable energy access to power grid in recent years, this review also analyzes the challenges brought by it and provides some corresponding suggestions. It is hoped to provide some guidance for the subsequent research work.

1. Introduction

With the continuous expansion of power demand and the continuous support of governments all over the world, the capacity and scale of power system show a trend of rapid growth. At the same time, the requirements for the safe and stable operation of the system have also been raised to a new height. Although a large number of control equipment and a variety of automation devices are installed in the system, the security and stability of the system is still seriously challenged [1–3].

Oscillation is such a common public hazard in power system, where the out-of-step oscillation is one of the most serious phenomena, belonging to the system level [4,5]. It is caused by system power imbalance, characterized by the loss of synchronization of generator sets, rapid changes in potential frequency, severe fluctuations in network voltage and current, and frequent power interactions between subsystems, which can cause equipment damage, or cause large-scale power outages [6]. After its occurrence, the system must be quickly and reasonably divided into several isolated islands which can operate independently and stably under a series of control strategies. Out-of-step splitting control is such a control mean, which is takes as an important part of the “third line of defense” of power grid protection system. It must reasonably solve the following three problems [7]:

- (1) When should the system be split reasonably?
- (2) Where are the optimal splitting section?
- (3) How can maintain the stability of the islands after splitting and achieve reconnect?

However, it is highly challenging because the determination should be made from a number of electrical components to meet many operational constraints in each island, but the allowed time is short (realtime, only a few seconds) [8,9]. Optimality and rapidity are a contradiction. If the judgment of out-of-step oscillation state is not accurate or the decision-making is late and not properly, it would directly lead to the failure of prevention and control, the out-of-step oscillation will spread to the whole system or evolve into other oscillation modes, eventually leading to the collapse of the system. In the past 30 years, major blackouts relative with it always occurs worldwide, the impact on the society and economy is serious [10]. The famous case is the North America power grid blackout occurred on August 14. Because of the failure of out-of-step oscillation identification, the instability range continued to expand. After the out-of-step oscillation, the generators jumped because of protection, the branch tripped due to the chain reaction, eventually causing energy loss of 61,800 MW and economic loss of 6.4 Billion dollars [11,12]. On February 2011 in Brazil, the out-of-step protection of one substation malfunctioned after out-of-step occurs, finally causing

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the instability of the northeast area grid in this nation, the power outage range covers seven states [13]. On July 2012 in Indian, the power grid was quickly split after out-of-step oscillation occurred. But the selection of splitting section is unreasonable, large amount of unbalanced power existed in split islands, causing the system further collapsed [14,15]. On March 21, 2018, in the out-of-step accident of the Northern and Northeastern power grids of Brazil, the installed out-of-step protections refused to operate because the out-of-step centers constantly migrated in the system, the security of the system suffered a major threat [16]. Obviously, the existing splitting control methods is facing major challenges from the fast and changeable operation mode and system architecture of the complex large power grid.

Besides, under the guidance of national strategies such as “Carbon Peak” and “Carbon Neutrality”, large amount of photovoltaic, wind power and other renewable energy generation equipment are continuously connected to the power system. At the beginning of 2020, the proportions of renewable energy generator capacity in many regions have exceeded 30%. It is estimated that wind and solar generation would be up to 50% of world electricity in 2050. The power grid is undergoing major changes quietly [17,18]. However, such a large amount of low-inertia capacity addition would increase the unpredictability in grid operation, the out-of-step mode changes more rapidly and easily evolves into the multi-mode oscillation, the locations where the system would split during an out-of-step condition may change, the generic out-of-step splitting control faces the unprecedented challenges. This puts forward higher requirements for the optimality and rapidity of out-of-step splitting control, and it is urgent to carry out principle innovation and technical route innovation on the basis of the original method.

In the current research stage, how should we continue our research to develop a more a highly reliable, adaptive, and ultra-fast out-of-step splitting control strategy? How to overcome the shortcomings of the existing out-of-step splitting control methods in complex power grids and the new challenges brought by large-scale renewable energy integration? This has attracted much attention. Under this background, this review is written to track the development of out-of-step oscillation splitting control, analyzes their technical routes and give some suggestions. It is hoped to provide some guidance for the subsequent research work.

The rest of the paper is organized as follows. In Section 2, the detailed introduction to the concept of out-of-step oscillation is given. In Section 3, the out-of-step splitting control technologies is combed according to the development sequence of three stages, and their shortcomings are summarized through detail analyses and comparison. In Section 4, the challenges brought by the large-scale access of renewable energy are analyzed, and some relevant future research suggestions are given. It is hoped to provide some guidance for the subsequent research work.

2. Basic concept of out-of-step oscillation

(i) **Definition of out-of-step oscillation.** Out-of-step oscillation is a serious oscillation phenomenon with frequent power interaction among subsystems, it belongs to the low frequency oscillation category. After the system is disturbed, large amount of imbalance power would exist in the system. Under this action, the generator rotors would accelerate or decelerate in different level, finally the synchronization between generators might lose, the out-of-step oscillation occurs at this time [19]. In general, when the potential phase angle difference δ between the two terminal equivalent systems $\dot{E}_M$ and $\dot{E}_N$ in Fig. 1 exceeds 180° , it is considered out-of-step oscillation occurs in system. Usually, when $\delta > 120^\circ$, the out-of-step splitting control equipment starts. According to statistics, the frequency slip of the system is usually $1 \sim 7$ Hz, the first out-of-step oscillation cycle is about 1s [20].

(ii) **Variation characteristics of system electrical quantity.** During the out-of-step oscillation process, δ periodically change within the

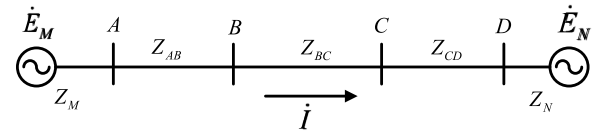


Fig. 1. Equivalent two-terminal power system.

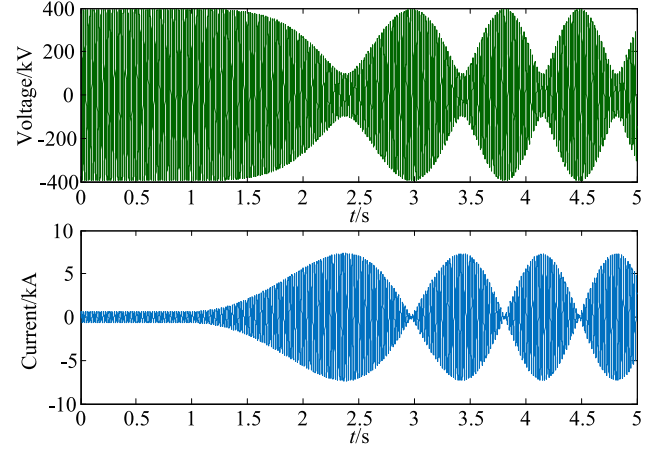


Fig. 2. Voltage and current variation in the out-of-step oscillation.

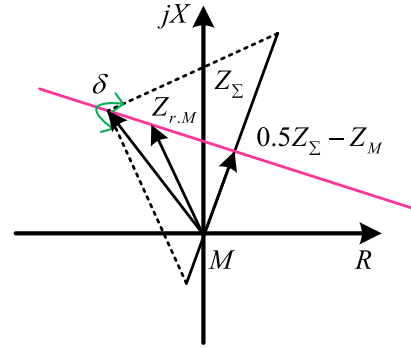


Fig. 3. Moving trajectory of measurement impedance.

range of $0^\circ \sim 360^\circ$, the real-time variations of system current and bus voltage are displayed in Fig. 2 shows, they satisfy (1) and (2) [21,22]

$$I_M = \frac{E_M}{|Z_\Sigma|} \sqrt{1 + h^2 - 2h \cos \delta} \quad (1)$$

$$U_M = E_M \sqrt{h^2 + (1 - m)^2 + 2h(1 - m) \cos \delta} \quad (2)$$

where $\dot{E}_M = E_M$, $\dot{E}_N = E_N e^{-j\delta}$, $h = \frac{E_N}{E_M}$, $Z_\Sigma = Z_M + Z_{AB} + Z_{BC} + Z_{CD} + Z_N$. When $\delta \approx 180^\circ$, the amplitude U_M of bus voltage $\dot{U}_M$ obtains its minimum value, and when $\delta \approx 360^\circ$, U_M obtains its maximum value. However, for the system current $\dot{I}_M$, the situation is opposite. Besides, the measurement impedance satisfies

$$Z_{r.M} = \frac{1}{1 - h e^{-j\delta}} Z_\Sigma - Z_M \quad (3)$$

Its moving trajectory is shown in Fig. 3. It slowly moves through the quadrant 1 to the quadrant 2 of coordinate system, and obtains its minimum amplitude value when $\delta \approx 180^\circ$. The above three laws are the basic principles of the local measurement based out-of-step protection.

(iii) **Definition of out-of-step oscillation center and out-of-step center.** The oscillation center is a special point in the out-of-step oscillation system, defined as the point whose voltage amplitude is the minimum in the system at a certain moment. Taking the equivalent

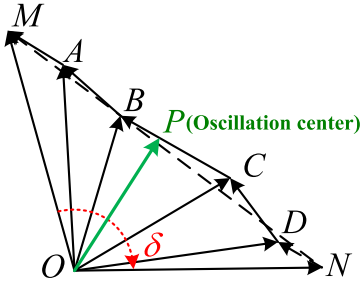


Fig. 4. Relationship between phasors in the equivalent two terminal system.

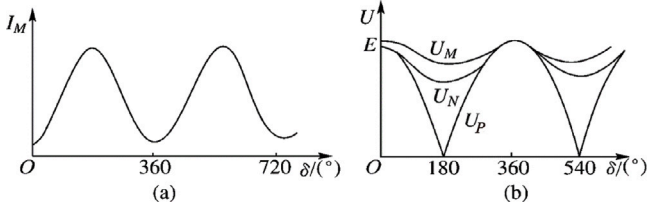


Fig. 5. Variation of current amplitude and bus voltage.

two-terminal system in Fig. 1 as an example, the phase relationships between voltage phasors are shown in Fig. 4. For a point P in the system, its voltage $\dot{U}_P$ can be respected by the phasor $\overline{OP}$ connecting the point P on the poly-line $MABCDN$ and the coordinate origin O , $|OP|$ indicates the amplitude of $\dot{U}_P$. At any time, there is a point in system whose length is the shortest among all $\overline{OP}$, this point is the oscillation center [23].

The out-of-step center is a special out-of-step oscillation point, which has important significance for the instability mode of system. For a complex system with unequal potentials and different impedance angles of the entire system, the position of oscillation center will continue to migrate as the oscillation progresses, and its corresponding voltage amplitude will also change periodically like $\dot{U}_P$ in Fig. 5. When $\delta \approx 180^\circ$, the voltage amplitude of oscillation center would decrease to zero. This oscillation center is usually called as out-of-step center [24].

3. Development of out-of-step splitting control

3.1. Three stages of development of out-of-step splitting control

The research about the out-of-step oscillation splitting protection and control has gone through a long time. Two major changes have taken place due to the influence of the WAMS system and the large-scale integration of renewable energy sources, making it go through three development stages like Fig. 6 shows.

Stage 1: Local measurement information based out-of-step splitting control for traditional complex power system. In the initial development stage, the speed and accuracy of measurement and synchronous communication are far less developed than they are now. Designers can only use local measurement information to design out-of-step splitting protection. Its main task is to quickly and accurately implement splitting after detecting out-of-step oscillations to prevent the spread of oscillations. The research started in the 1980s [25,26] and are conducted by the Academician Guorong Shen of Chinese Academy of Engineering, academician Jiali He of Russian Academy of Engineering, etc. RCS-993 A out-of-step splitting equivalent is a classic product of this stage, and it is still in service.

Stage 2: Wide-area measurement information based out-of-step splitting control for traditional complex power system. Since the new century, large number of wide-area phasor measurement units (PMU) have been installed in the power system, the electrical information on the nodes in the whole system can be obtained real-time, which

provides a good platform for studying the new class of out-of-step splitting control [27–29]. How to achieve better splitting is reconsidered. In this stage, the dynamic monitoring type and the dynamic optimization type wide-area information out-of-step splitting control methods are proposed. According to different principles, the latter can be further divided into the slow coherency theory type, graph theory type and dynamic optimization type.

Stage 3: Wide-area measurement information based out-of-step splitting control for renewable energy system. The large-scale access of renewable energy sources accelerates the evolution of out-of-step oscillation modes and presents different characteristics. This has triggered scholars to comprehensively think about this, but it is still in its infancy.

3.2. Constraints section of out-of-step splitting control

Out-of-step splitting control is directly related to the safe and stable operation of the system after splitting. The rapidity and optimality of out-of-step splitting determination are the two core problem to be solved in out-of-step splitting control (their relationship is shown in Fig. 7), which are accompanied by all the three stages of the development of out-of-step splitting control.

Optimality: In order to select an optimal out-of-step splitting section to ensure the stability of the island after splitting, it should ensure the objective function such as Power Imbalance (PI, Eq. (4)) or Power Flow Disruption (PFD, EQ. (5)) minimal, and also should consider the aspects of island voltage stability, the generator coherency, and the generation-load power balance, transient stability, etc. Through literature review and induction, these constraints are summarized and listed in Table 1. Obviously, the out-of-step splitting control problem is a multi-objective and multi-constraint problem.

$$\min PI = \min \left| \sum_{\substack{I:(i,j) \in E \\ i \in V_1, j \in V_2}} P_l \right| \quad (4)$$

$$\min PFD = \min \sum_{\substack{I:(i,j) \in E \\ i \in V_1, j \in V_2}} |P_l| \quad (5)$$

Rapidity: With the system structure continues to become more complex, more and more nodes and instability modes are included in the selection of splitting section. It no doubt increases the difficulty of the out-of-step splitting section determination. Just for the IEEE 118 system, the existing 186 lines create 2186 possible choices for system splitting. Determining the out-of-step splitting section is an extremely complex and time-consuming process. However, the national grid of China requires the out-of-step system should be split within 1~2 out-of-step oscillation period. According to the statistics, the oscillation period is usually 1 Hz to 7 Hz. It leaves us only 0.28s~2s to process. Obviously, the out-of-step splitting is also a narrow timescale decision-making process.

It can be seen that the contradiction between the optimality and rapidity of out-of-step splitting control is prominent. The search strategy can ensure all the constraints are satisfied, and also meet the rapidity requirement does not exist, unless the program processor has powerful computing capabilities. Obviously, it is impossible in the actual working environment of protection. In fact, at different development stages of out-of-step splitting control, the trade-off between optimality and quickness is made according to the design requirements and design objectives. Therefore, constraints are trade-offs and only a small sub-set is considered, such as the frequently used combinations of load-generation balance and generator coherency. According to the arrangement of the article, this will be sorted out in detail in the following part.

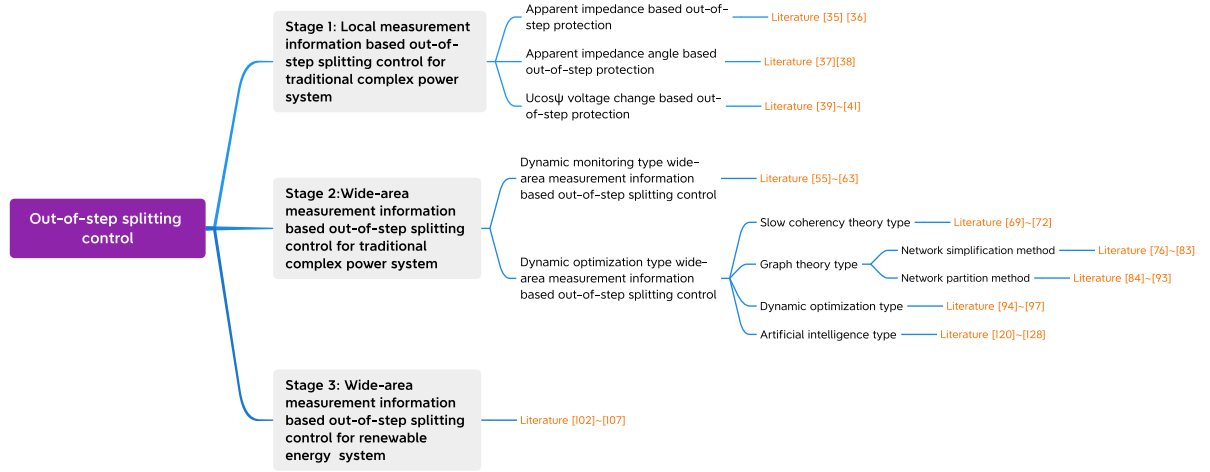


Fig. 6. Classification of out-of-step splitting control methods.

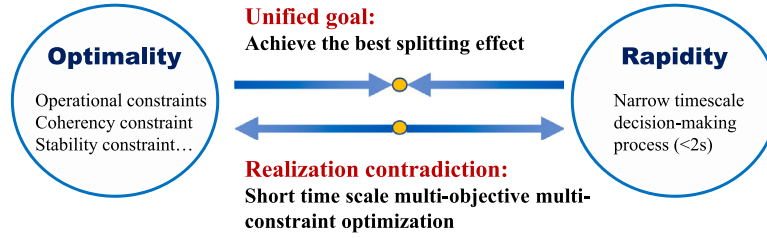


Fig. 7. Requirements and problems of out-of-step splitting control methods.

Table 1
Constraints of out-of-step splitting control.

Constrain name	Constrain description	Constrain formula
Partitioning constraint	Every node belong to exactly one island. A line is in the final cutset if neither of the ends of the line lies in the same partition.	
Connectivity constraint	All buses of every island, including generator and load buses, are connected. No isolated nodes and unstable islands.	
Operational constraints	(1) The output of generators should be within the range of their maximum and minimum outputs; besides, the up and down power outputs of generators in a short time are also limited. (2) Due to the thermal stability, voltage stability and economic concerns, the power flow in transmission lines, bus voltage, and switch number of transmission lines are limited.	$\begin{cases} P_{\text{gen},i}^{\min} \leq P_{\text{gen},i} + P_{\text{up},i} - P_{\text{down},i} \leq P_{\text{gen},i}^{\max} \\ Q_{\text{gen},i}^{\min} \leq Q_{\text{gen},i} + Q_{\text{up},i} - Q_{\text{down},i} \leq Q_{\text{gen},i}^{\max} \\ 0 \leq P_{\text{up},i} \leq P_{\text{up},i}^{\max} \\ 0 \leq P_{\text{down},i} \leq P_{\text{down},i}^{\max} \\ 0 \leq Q_{\text{up},i} \leq Q_{\text{up},i}^{\max} \\ 0 \leq Q_{\text{down},i} \leq Q_{\text{down},i}^{\max} \end{cases}$ $\begin{cases} -y_{ij} P_{ij}^{\max} \leq P_{ij} \leq y_{ij} P_{ij}^{\max} \\ -y_{ij} Q_{ij}^{\max} \leq Q_{ij} \leq y_{ij} Q_{ij}^{\max} \\ V_i^{\min} \leq V_i \leq V_i^{\max} \\ N_{\text{line}} - N_{\text{switch}}^{\max} \leq \sum_{i=1}^{N_{\text{bus}}} \sum_{j=1}^{N_{\text{bus}}} y_{ij} / 2 \end{cases}$
Frequency stability constraint	the power imbalance on the created island must not exceed a certain value.	$\Delta P_{\text{imb}}^k \leq \frac{2H_k}{T_0} (df/dt)_{\max} \quad k \in K$
Generator coherency constraint	All coherent generators must remain in the same partition during ICI to avoid transient instability of the islanding operation.	$\begin{cases} G_i^m \subset V_i^G \quad \text{where } G_i^m \text{ own same coherence} \\ \sum_{i=1}^k V_i^G = V^G \\ V_i^G \cap V_j^G = \emptyset \quad \forall i \neq j \end{cases}$
Reactive Power Regulation Constraints	SVC and CBs are widely deployed for reactive power compensation. SVC can output or absorb reactive power continuously while CBs can output reactive power discretely only.	$\begin{cases} Q_{\text{SVC},i}^{\min} \leq Q_{\text{SVC},i} \leq Q_{\text{SVC},i}^{\max} \\ Q_{\text{CB},i} = N_{\text{CB},i} Q_{\text{CB},i}^{\text{each}} \\ 0 \leq N_{\text{CB},i} \leq N_{\text{CB},i}^{\max} \quad N_{\text{CB},i} \in \mathbf{Z} \end{cases}$

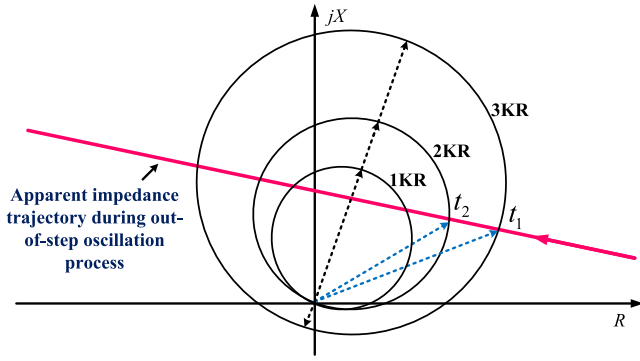


Fig. 8. Apparent impedance based out-of-step protection principle.

3.3. Method combining on three development stages of out-of-step splitting control

3.3.1. Local measurement information based out-of-step splitting control for traditional complex power system

The traditional local measurement information based out-of-step splitting control adopts the working mode of “offline setting + online action”. This kind of method off-line calculates the out-of-step oscillation modes and the splitting points according to the operation state and architecture of power grid, and designs different splitting methods and threshold values according to the different electric quantity change characteristics in the out-of-step oscillation process. In the actual operation, once the preset constraints are satisfied, the equipment performs splitting operation. At the same time, if the system structure or operation mode changes, it is likely to cause the protection to refuse to act.

There are three representative local information based out-of-step splitting control schemes, including the apparent impedance based method, the apparent impedance angle based method, and the $U_{\cos\varphi}$ voltage change based method. Below, their principles would be introduced.

(i) **Apparent impedance based out-of-step protection.** It utilizes the apparent impedance trajectory to distinguish between short circuit and out-of-step oscillation [30,31]. Its working principle is shown in Fig. 8. When out-of-step oscillation occurs in system, the apparent impedance gradually varies, it enters 3KR at t_1 firstly, then 2KR at t_2 and 1KR lastly. However, when short circuit occurs and it is in the first-zone unit 1KR of distance protection, the apparent impedance abruptly changes from the load impedance to the short-circuit impedance, the first-zone unit 1KR, second-zone unit 2KR and third-zone unit 3KR of distance protection operate at the same time. Therefore, if the time duration Δt between t_2 and t_1 is greater than the setting value t_{set} , an out-of-step oscillation can be judged. This is the basic principle of this type of methods, many similar methods are derived accordingly.

(ii) **Apparent impedance angle based out-of-step protection.** It utilizes the time series of apparent impedance angle variation to detect out-of-step oscillation [32,33]. It divides the apparent impedance angle area into six parts like Fig. 9 shows. After out-of-step oscillation occurs, if the oscillation center locates at the positive direction of protection, the apparent impedance angle would pass through the sections I, II, III and IV successively. When short circuit fault occurs within the scope of protection, the apparent impedance angle directly changes from I to II and remains in this state. Through observing the time series of apparent impedance angle variation, the out-of-step oscillation can be determined.

(iii) **$U_{\cos\varphi}$ voltage change based out-of-step protection.** It uses the calculated value by multiplying the measured voltage amplitude U at bus and the phasor angle difference φ between voltage and current to indirectly reflect the voltage amplitude change of oscillation center [34–36]. Its variation regulation is shown in Fig. 10. It could be

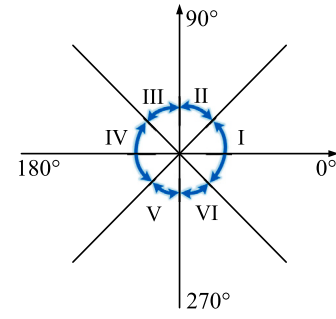


Fig. 9. Apparent impedance angle based out-of-step protection principle.

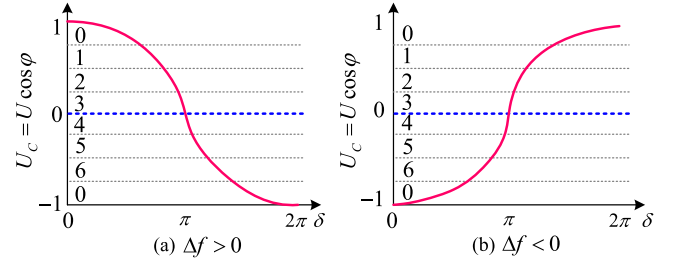


Fig. 10. $U_{\cos\varphi}$ voltage change based on out-of-step protection.

found that the voltage $U_{\cos\varphi}$ continuously change with δ in the out-of-step oscillation process and crosses the 0 axis at $\delta \approx 180^\circ$. While a short circuit occurs, $U_{\cos\varphi}$ directly decrease to 0, there is no any transition process. Therefore, through observing the time series changes of $U_{\cos\varphi}$, the out-of-step oscillation can be quickly identified.

3.3.2. Wide-area measurement information based out-of-step splitting control for traditional complex power system

As an important technical support for smart grids, the Wide Area Measurement System (WAMS) provides a new platform and great flexibility to solve the out-of-step splitting control problems [37]. As Fig. 11 displays, WAMS real-time obtains the electrical measurement information of all important nodes in the system, and sticks the time stamp based on the high-precision timing signal sent by GPS for synchronization, finally shares them in the whole network [38–40]. Until now, its refresh rate can be up to 100 frame/s for 50 Hz system, 120 frame/s for 60 Hz system (see Table 2) [41,42]. Due to geographical distribution, the transmission of PMU data to the Phasor Data Concentrators (PDC) has a transmission delay, and the transmission time is different. However, according to statistics, the data latency with optical fiber transmission network can be limited to the order of 50 ms [43]. As long as the configurable wait time during which the measurements are transmitted to the application is set reasonable (< 200 ms), the communication delay problem can be ignored. On the other hand, from the coverage aspect, the WAMS systems have been widely equipment in the countries of Europe, North America, China and Russia, such as the WSCC system in North America, CSE in Spain, etc [44–46]. According to “China Southern Power Grid Synchronized Phasor Measurement Unit Specification”, PMU configuration should meet the requirements of grid stability monitoring, analysis, early warning and decision-making, requires that the power plant with single generator capacity 300 MW and above, power plant with a total installed capacity of 500 MW and above, the 220 kV substations exceeding 200 MW should be equipped with PMU devices, other power plants and substations with outstanding stability problems should also be equipped. Besides, US DOE reports the PMUs in North America have increased by an order of magnitude in the past ten years and still show a sustained growth trend. The condition for using wide-area information on the online out-of-step splitting control

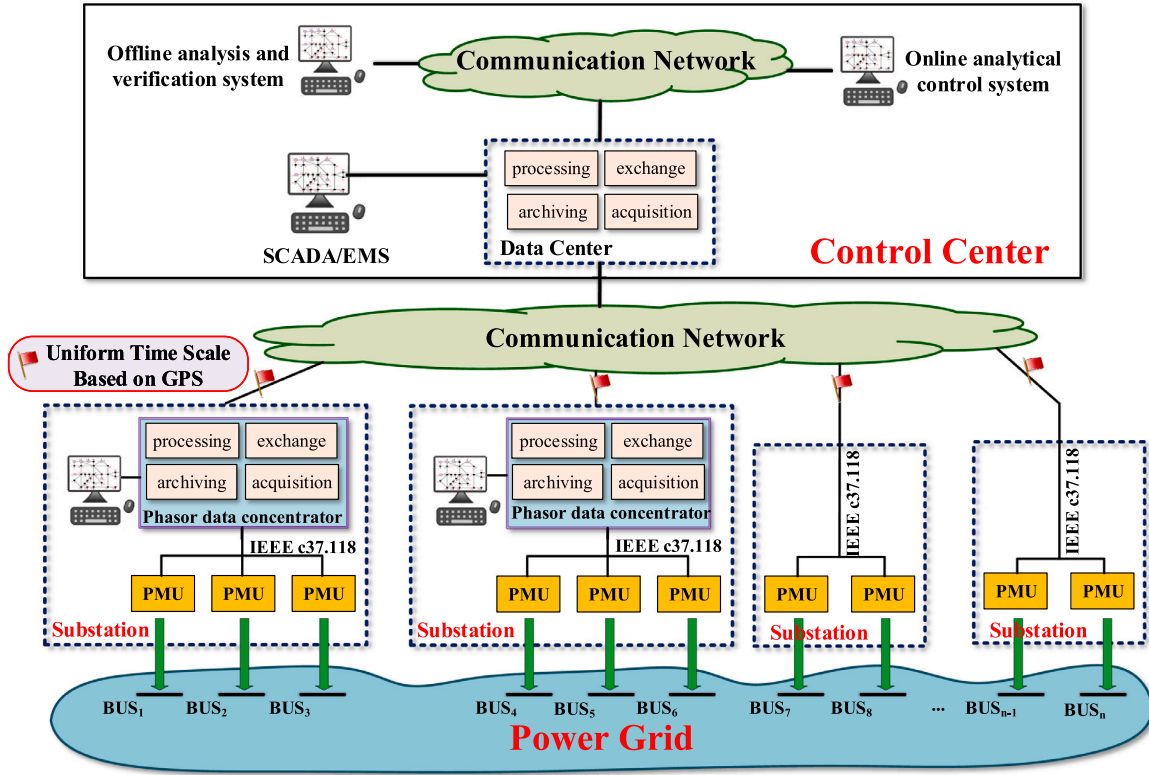


Fig. 11. Wide area measurement system.

Table 2
Required PMU reporting rates.

System frequency	Reporting rates (frames per second)							
50 Hz	10	25	50	100				
60 Hz	10	12	15	20	30	60	120	

are more and more mature. It provides to the power system operators a real-time wide area visualization of the system operating condition.

Under this background, the research work on the wide area information based out-of-step splitting control has begun, and a lot of fruitful research results. According to the difference in terms of control routes, these methods can be further roughly divided into the dynamic monitoring type and the dynamic optimization two sub-classes. Below, they would be detailed instructed respectively.

(i) Dynamic monitoring type wide-area measurement information based out-of-step splitting control.

This sub-class of methods uses the working mode of “online monitoring the voltage and current changes + dynamic determining the splitting section and implementing splitting”. They inherit the splitting ideas of the traditional out-of-step splitting control strategy, still considers the out-of-step centers represent the instability mode and takes the as the splitting points. But the difference is this sub-class of methods uses the wide-area information provided by WAMS dynamically to determine the locations of out-of-step centers and implement the splitting operation on these points. It is no longer limited by the changes of system structure and operating modes like the traditional out-of-step splitting methods. Achieving rapid splitting is the goal pursued by this sub-class of methods.

Literature [47] utilizes the stability-preserving transformation to obtain the power angle motion trajectory $p(\delta)$ of the imaging single-machine system, and based on the extended equal-area rule quantization analysis technique to judge out-of-step oscillation when the dynamic saddle point appears. Literature [48] detects the out-of-step oscillation by the aid of the motion trajectory of rotational speed

difference-power angle difference $\Delta\omega - \Delta\delta$ on the phase plane. In order to accelerate the judgment, an out-of-step oscillation prediction method is proposed in literature [49], which predicts the power and rotor angle curves by auto-regressive moving average method (ARMA). In literature [50], based on the inertial center concept, an intelligent observer is designed for identifying the out-of-step by extending the velocity–acceleration trajectory curve. It improves the out-of-step decision speed. Our research team also make some efforts in this research area [51–53]. Specifically, we studied the affecting mechanism of the system structure and operation mode changes on the location of out-of-step center by constructing the position function of system nodes. In addition, we study the phase relationship between voltage and current during out-of-step oscillation, and develops a fast oscillation center tracking and out-of-step center positioning algorithm. In literature [54], the variation characteristics of bus voltage frequency during the out-of-step oscillation process is analyzed, and a bus voltage frequency based out-of-step center locating method is proposed. In literature [55], the expression of measured impedance in the multi-frequency oscillation scenario is deduced and its variation is analyzed. After defining the position factor of out-of-step center, a novel out-of-step center locating method suitable for multi-frequency oscillation scene is proposed.

(ii) Dynamic optimization type wide-area measurement information based out-of-step splitting control.

This type of method regards the splitting control problem as an optimization problem, that is, how to select the splitting section that satisfies the constraints in a huge power grid. Optimality is the biggest feature of stupid methods. For determining the optimal out-of-step splitting section, a series of methods have been proposed. According to different principles, it can be divided into the slow coherency theory type, graph theory type and dynamic optimization type methods. A detailed introduction is given below.

A. Slow coherency theory type.

Slow coherence theory is a tool that comprehensively considers multiple time scales and singular perturbation characteristics to study slow coherence phenomena in power systems. By solving the eigenvalues

and eigenvectors of the state matrix of the dynamic equation of the power system, the theory analyzes the existing oscillation modes of the system, and obtains the weak connection between the generator groups, so as to search for the solution section in the weak connection area. This method can analyze the coherence of the generator sets, and the splitting section search results have nothing to do with the fault type, fault location and other information, which can effectively reduce the scale of the splitting decision space.

In Ref. [56], the mathematical mechanism and conditions of different out-of-step oscillation modes in power system are explained by means of slow invariant manifold theory on singularly perturbed model; After defining the weak connection of power system, the rationality of splitting at the weak connection of power system is explained theoretically when the system is out-of-step. Literature [57] studied the dynamic characteristics of the power system and found that the system behaves in two time scales, fast and slow, and proposed a method for clustering generators based on the slow coherence theory and determining the weak connection area of the system according to the eigenvalues of the dynamic equation method. Literature [58] solves the slow coherence problem of load nodes after introducing the generalized characteristic analysis method, and completes the slow coherence area division of system according to the system topology. Literature [59] introduces the load nodes into the slow coherence characteristic matrix, and simultaneously group the generators and load nodes into coherent groups. Ref. [60] first determine the number of coherent generators in the system and the coherent generators are equivalent to virtual generator nodes. After introducing the weight index to measure the “distance” between adjacent nodes, the improved Laplace partition strategy is used to heuristic neighborhood search for the optimal out-of-step splitting section.

Using the slow coherency theory can reduce the search range and search time of the optimal out-of-step splitting section, but this out-of-step protection method relies too much on the slow coherency characteristics of the generator set during the steady-state operation of the system. In the actual operation process of power grid, its operation state is usually much more complex. On the one hand, the topology and operation mode of complex interconnected power grid usually change according to the actual operation needs. Specially, the out-of-step oscillation often occurs after complex cascading failures, both the topology and power flow of the system will change. In practice, the results of generator coherency grouping may be inconsistent with the slow coherency characteristics determined in steady-state operation. Using the slow coherency characteristic of the generator in the steady state of the original system to determine the out-of-step section of the current system will bring unreasonable solution.

B. Graph theory type.

Graph theory is an efficient searching method for reducing the searching space of splitting scenarios. When it is utilized to find the optimal splitting section that separates coherent generator groups with operational constraints, the determination process is transformed into a graph optimal cutting (GOC) problem. It was proved it is a NP problem because the search space of out-of-step splitting section increases exponentially with the increases of the nodes and complexity of power network [61,62]. It cannot be solved in polynomial time. In order to improve the search efficiency, the network simplification methods and network partition methods are proposed.

1. Network simplification method.

The basic idea of the network simplification method is to aggregate the network nodes and simplify the network structure with the help of graph theory technologies after knowing the coherence information of the generator after out-of-step oscillation, finally improving the search efficiency of the optimal out-of-step splitting section. Refs. [63–65] utilize the power grid characteristics and graph theory to pre-treatment the power network and simplify it to the one can be searched for out-of-step splitting section; then use the ordered binary decision diagram (OBDD) algorithm to search the set of splitting sections meeting the

pre-set constraints; finally determine the most reasonable out-of-step splitting scheme by comparison. It contributes to represent the pre-set constraints into an explicit format by Boolean variables. But it also leads to it must bear the cost of high-intensity computation. For satisfying the real-time application computational needs, the network model must be simplified to contain less than approximately 40 nodes [66]. Literature [67,68] propose the out-of-step splitting section search methods with multi-layer graph segmentation theory. These methods first coarsen the power system diagram according to the known coherence group results of out-of-step generators, then uses the greedy graph growth algorithm to complete the initial area division, and considers the topological constraints to finally optimize the out-of-step splitting section. Literature [69] uses the knapsack problem to establish the optimal graph decomposition model and searches the optimal out-of-step splitting section in two steps of “search + adjustment”. Besides, the authors use the graph knapsack technology to convert the out-of-step splitting into the master-slave problem, and search the optimal out-of-step splitting scheme through the alternate iterations between master and slave problems [70].

2. Network partition method.

Different from the network simplification method, the fast network partition method does not simplify the system, but directly divides the network from the operation characteristics of power system. The most representative are the spectral clustering methods and their derivative methods. Literature [71–73] firstly determine the coherent generator groups based on the dynamic model of the generators, and then select the electrical components need to be split with the spectral clustering method to minimize power flow disruption and meet the restrictions of generator coherency. However, its intensive computation is hard. A novel constrained spectral clustering approach is offered in [74,75], to reduce the computational complexity and time. In [76], the hierarchical spectral clustering algorithm is applied. Even through more power system islanding constraints are introduced, the generator coherency is not considered. To obtain more sustainable islands, literature [77] employed a multi-layer clustering approach to minimize the reactive power disruption as well as the active power. Besides, literature [78] uses the current tracking method to partition the system, and adjusts the island with the goal of reducing the unbalanced power in the island as much as possible. Literature [79] considers the effect of node voltage control ability and active power balance, and proposes a composite electrical distance to measure the closeness of the connection between nodes; then the k -medoids clustering algorithm is used to determine the system partition scheme. A multilevel kernel k -means algorithm [80] is proposed to obtain the islanding scheme by bus aggregation, graph partition, and rejoining.

C. Dynamic optimization type.

In this type, the out-of-step splitting control is solved through mathematical programming of the desired objective function and corresponding constraints. In literature [81], the generator similarity symmetric matrix called D_g is proposed to identify coherent generator groups, and the binary linear programming (BLP) base algorithm is designed to search the optimal splitting section that minimizes the disruption of the power transferred between these islands. In the whole operation process, there are a lot of matrix operations. In literature [82], the splitting section searching work is expressed as a mixed-integer formulation with considering the slow coherency of synchronous generators and minimum load shedding. It is solved using Benders Decomposition (BD) technique. Literature [83] points out the above strategy belongs to a DC-based MILP formulation. It cannot guarantee the AC constraints of the network. So, it builds a linear AC load flow model and considers the frequency stability constraints of resulted islands. The developed MILP model including the discretized system frequency responses have been coded in GAMS and solved using CPLEX solver. The base of this algorithm is a MILP formulation which is of exponential complexity; hence its execution time is also exponential. In literature [84], a search space reduction strategy is proposed, which generates the additional

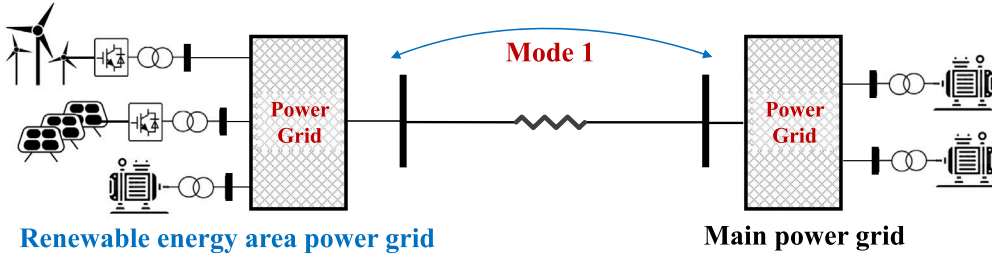


Fig. 12. Out-of-step oscillation mode between the renewable energy area power grid and the main power grid.

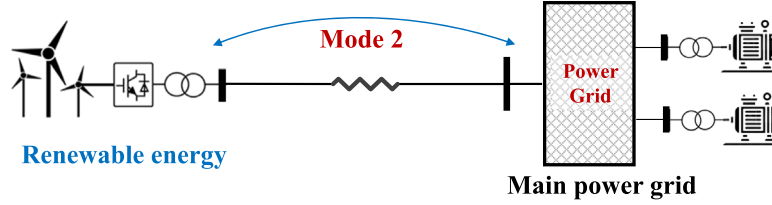


Fig. 13. Out-of-step oscillation mode between the renewable energy and the main power grid.

constraints to reduce the search space and shorten solution time. The literature [85] further optimizes the above methods and further expands the scope of constraints. A frequency stability constraint as well as an indicator of risk of voltage instability are adopted to improve both voltage and frequency stabilities, meanwhile, the power balance and operational limit constraints are considered. Based on this, a multi-stage Linear Programming (LP) algorithm is designed. Literature [86] involves the uncertainties of renewable energy sources into the splitting section determination. Besides, a Fuzzy C-Means (FCM) clustering method with F-statistics is utilized to identify the coherent generator groups with the frequency data measured by Phasor Measurement Units and a Adjustable Robust Optimization Programming (AROP) is designed considering coherent group constraints, connectivity constraints and robustness constraints about RES are presented.

3.3.3. Wide-area measurement information based out-of-step splitting control for renewable energy system

In the above two stage, the out-of-step splitting control are all proposed based on the traditional AC power system. However, since the beginning of the new century, numerous renewable energy have been continuously connected to the grid, which makes a huge impact on the causes of out-of-step oscillation, the influence scope and development speed of out-of-step oscillation, as well as the corresponding splitting control schemes. In details,

(i) The causes of out-of-step oscillation change. In the traditional power system, the causes of out-of-step oscillation is the large power disturbances, such as short circuit, generator shutdown, load shedding and etc. Under the action of unbalanced power, the synchronous generator rotor accelerates or decelerates. If the power angle difference between different generator sets continues increasing up to 180 degrees, it can be determined the system out-of-step oscillates [87]. In the renewable energy system, there are two modes of out-of-step oscillation, including the out-of-step oscillation mode between the renewable energy area power grid and the main power grid (named as Mode 1, like Fig. 12) [88], and the out-of-step oscillation mode between renewable energy and the main power grid (named as Mode 2, Fig. 13) [89]. For Mode 1, the causes are consistent with that of the out-of-step oscillation in traditional power system. But the disturbance is relatively small. The causes for Mode 2 mainly include two [90]. One cause is short circuit. Under this condition, the ability of main power grid to receive renewable energy power decreases, resulting in the mismatch with the renewable energy generation power reference and power transmission. Then, the virtual power angle of renewable

energy power generation equipment increases, which finally leads to out-of-step oscillation. The other cause is the weak network condition, which usually appears at the end of power grid. Affected by the weak network condition, if the active power reference of renewable energy power generation exceeds the line power acceptance capacity, the virtual power angle of renewable energy power generation equipment also increases, finally leading to out-of-step oscillation.

(ii) The influence scope and development speed of out-of-step oscillation change. Out-of-step oscillation in the traditional power system is a system-level oscillation phenomenon. The influence scope is large, reaching up to the entire power grid. The consequences are very serious. In the renewable energy system, due to the large-scale connection of renewable energy, the renewable energy area power grid's inertia reduces, the disturbance that triggers out-of-step oscillation in Mode 1 is smaller than that triggers the out-of-step oscillation in traditional power system, the development speed is quicker, and the occurring frequency is also higher [91]. However, compared with Mode 1, renewable energy has smaller inertia relative to the renewable energy area power grid, the out-of-step oscillation under Mode 2 develops faster, the trigger disturbance and influence scope are smaller. Further, the different renewable energy control strategies own different inertia and have important influence on the out-of-step oscillation development speed. Scholars conducted relevant analyses and made important scientific research progress. Literature [92] points out the synchronous instability of grid-following (GFL) type renewable energy is directly related to the dynamic characteristics of the phase-locked loop (PLL) module based active power control loop. Literature [93] reveals the synchronous instability of grid-forming (GFM) type renewable energy is closely related to the second-order system of the virtual inertia control based active power control loop. Literature [94] further analyzes the synchronous instability principles in the multiple GFL type renewable energy parallel system. Literature [95] further studies that of the GFL type renewable energy and GFM type renewable energy parallel system.

(iii) The out-of-step splitting control schemes change. In the traditional power system, it is usually necessary to quickly monitor whether the out-of-step oscillates in the system, search for the optimal out-of-step splitting section that satisfies the constraints, and implement splitting. In the out-of-step oscillation Mode 1 in renewable energy power system, the above work also needs to be done. But, less decision-making time is required for control because the development speed of out-of-step oscillation is accelerated [91]. In addition, the evolving form of out-of-step oscillation is more complex, causing the failure of traditional control methods. Literature [96] qualitatively analyzes

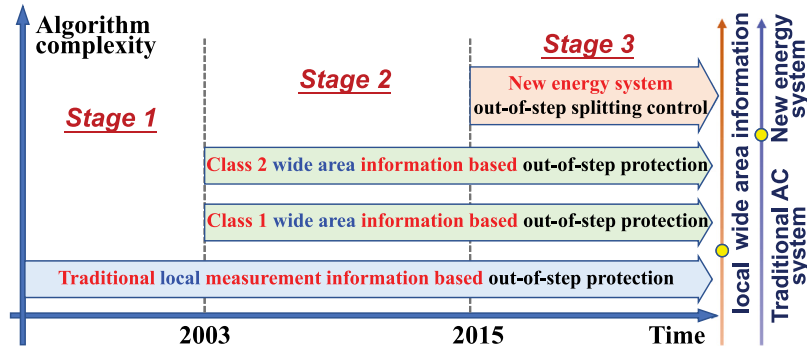


Fig. 14. Development of out-of-step splitting control method.

the influence of wind power access ratio, operation mode and reactive power compensation on the location migration of out-of-step center. In [97], an evaluation method on the influence of wind farm location on power system inter-regional oscillation dynamics is studied. Aboutaleb Haddadi of Montreal Institute of Technology, Canada, points out that wind power access would lead to the traditional impedance-based out-of-step protection rejection in literature [98], and gave simulation validation using the IEEE PSRC WGD6 model. In literature [99], the influence of photovoltaic power generation on the location distribution of out-of-step centers in multi-frequency oscillation scenarios is further analyzed. In literature [100], Ahmad Farid Abidin, Malaysia, visually describes the impact of photovoltaic power generation on the characteristics of electrical quantity changes and points out the large amount access of photovoltaic would also cause the traditional out-of-step protection refuse to operate. Under Mode 2, the out-of-step oscillation can be suppressed by adjusting the renewable energy source's own controls, such as reducing the active power input level [101] and increasing the line voltage [102,103]. In the worst case, the renewable energy power generation unit should be cut off directly.

At present, the research in this field is only at the initial stage. The deep-seated, systematic mechanism analysis needs to be further research, and the corresponding out-of-step splitting control suitable for the renewable energy power grid needs further study.

3.4. Technical route analysis and suggestions on the out-of-step splitting control

From the above analysis in Sections 3.3.1–3.3.3, we know, the research on the out-of-step splitting control have gone through three stages of development. It experiences the transformation of information selection from local measurement to wide area measurement, and the application scope from traditional AC power grid to renewable energy power system like Fig. 14 shows. In this process, the performance of these methods is improved continuously, but there are still some deficiencies needs to be improved, which are marked in blue in Table 3.

In order to clarify the tasks to be completed in the future, the technical routes of these methods would be sort out below.

At stage 1, due to the limitation of realistic conditions, only the local measurement is used in the traditional out-of-step splitting methods. Because they adopt the working mode of “offline setting + online action” (the splitting points are determined with large amount of offline calculations, different splitting equipment installed on these points coordinate to split online referring to the preset setting values), it has very fast decision-making speed. However, its splitting section and setting value are fixed, it has two major problems:

(1) It is difficult to adapt to the changes of system structure and operation mode.

(2) Split islands might further collapse because of the island power supply and demand imbalance.

They both belong to the problem of optimality and dynamic adaptability. It is for this reason that the reliability of this kind of methods

is faced with great challenges. For example, in the out-of-step accident of the Northern and Northeastern power grids of Brazil on March 21, 2018, the out-of-step centers are constantly migrating in the system, the out-of-step protections installed on 500 kV Gurupi-Serra da Mes, 500 kV Ribeiro Gonçalves-São João do Piauí, 500 kV Bom Jesus da Lapa-Igaporan III, etc refused to operate, the security of the system suffered a major threat [16]. Obviously, the existing local measurement based out-of-step splitting control cannot meet the requirements of out-of-step splitting of complex power system.

At stage 2, the wide-area measurement information starts to be used and two sub-classes of methods are formed.

The first sub-class can be regarded as an extension of the traditional out-of-step protection, it inherits rapidity of decision-making in traditional methods. But it changes to adopt the working mode of “online monitoring, online decision-making”. Due to the full use of wide-area information in the decision-making process, it can quickly detect the out-of-step oscillation and dynamically determine the position of out-of-step center according to the actual operating state of power grid. It solves the problem of the traditional out-of-step protection which is difficult to adapt to the changes of system instability modes. However, this sub-class of out-of-step splitting control still adopts the way of splitting the power system at out-of-step centers, the problem of the islands crash after splitting still exists because of the unbalance between the power supply and demand in islands.

The second sub-class of methods is complete different, it fully considers the generator coherency characteristics and the island power balance problem after the system out-of-step oscillates. The out-of-step section determined in this way will be more able to meet the requirements of power grid. Although they can be divided into slow coherency type, graph theory type and dynamic optimization type according to different technical routes, in fact, they adopt the same general idea. Like Fig. 15 shows, the whole process is divided into two steps. It first performs coherence identification on generators offline or online, and then based on the grouping results to search the splitting section satisfying the preset constraint conditions. It is a kind of splitting section searching strategy from outside to inside, the search scope covers the whole network, which makes it limited by the complexity of system structure and the increase of system nodes. The rapidity of this method is severely tested. Until now, it has not realized online application.

Obviously, even though two classes of wide-area information based out-of-step splitting control methods adopt different control routes and have their own advantages and disadvantages, they cannot effectively solve the two shortcomings of the traditional out-of-step splitting control at the same time, but they complement each other. If a fit point between them can be found and their advantages are combined to explore a new class of out-of-step splitting control, it would be of great practical significance.

At stage 3, under the new background of large-scale access of renewable energy, a new stage of research has begun. Until now,

Table 3

Comparison of traditional AC power network Class I and Class II wide area information based out-of-step splitting control methods.

Contrast Item Method		Oscillation Mode Detection			Splitting Section Search			Global Control
		Style	Speed	Accuracy	Style	Speed	Stability constraint	
Stage 1	Traditional local measurement based out-of-step protection	Fixed	Quick	Low	Fixed	Quick	No	No
Stage 2	Sub-class 1 wide area information based out-of-step protection	Dynamic	Quick	High	Dynamic	Quick	No	No
	Sub-class 2 wide area information based out-of-step protection	Dynamic	Slow	High	Dynamic	Slow	Cover	No
Stage 3	New energy system out-of-step splitting control	Qualitative analysis						

Note: The areas marked with blue in the table still need to be further improved

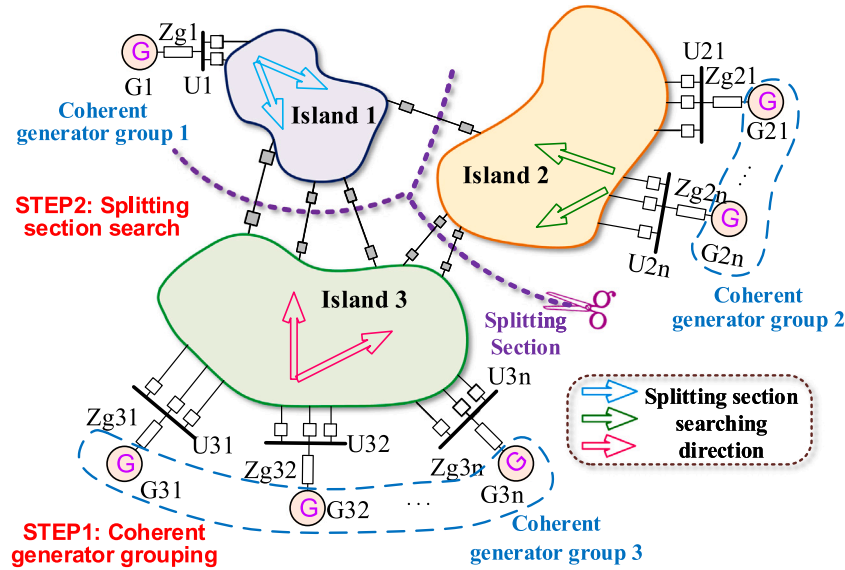


Fig. 15. Class II out-of-step splitting control method for traditional AC power grid based on wide-area information.

it is still limited to the qualitative analysis. the corresponding out-of-step splitting control has not been developed. But what is more noteworthy is that, with the development of power system and the realization of regional interconnection, the structure of power grid becomes more complex. The existing splitting control cannot be able to meet the splitting requirements of complex power grid. It is necessary to reconsider the splitting control problem from the perspective of the whole system. However, at present, the relative research on the stability evaluation and optimal control strategy for the out-of-step system is still lacking.

4. Some thoughts about the out-of-step splitting control after large-scale integration of renewable energy and AI technology applications

4.1. Challenges brought by large-scale renewable energy integration and current research status

In recent years, countries around the world have reached consensus on the development of renewable energy, and major strategies such as “carbon neutrality” have been introduced one after another. Large-scale renewable energy access to the grid becomes the norm for grid construction, which makes the power grid quietly undergone major changes.

For the out-of-step splitting control, it is a unprecedented challenge, mainly reflected in the following two aspects: (1) How to realize renewable energy power generation device self protection and power grid support under out-of-step situation? Renewable energy power generation device is essentially a kind of power conversion equipment driven by power electronics. It is easily destroyed due to the voltage and current fluctuations of out-of-step grid. It should first ensure the stability and safety of its own operation in the out-of-step oscillation condition. In addition, according to the requirements of the power grid, it also needs to provide necessary active and reactive power support and assist other protection controls in system to achieve oscillation suppression. At present, there is no systematic research on the above requirements. (2) How to deal with the severe requirements of large-scale access of renewable energy on reliability, rapidity and applicability of out-of-step splitting control protection? Renewable energy has inherent randomness, volatility, intermittence, small moment of inertia and deep coupling with power grid. Its large-scale access will aggravate the instability of power grid, showing the new characteristics of network complexity, instability polymorphism, evolution acceleration and so on. It puts forward more stringent requirements for the reliability, rapidity and applicability of out-of-step splitting control. The traditional Out-of-step splitting scheme is facing great challenges.

All in all, the large scale access of renewable energy generate the new and strict requirements on the renewable energy power generation device reliable ride through and the system global cooperative stability

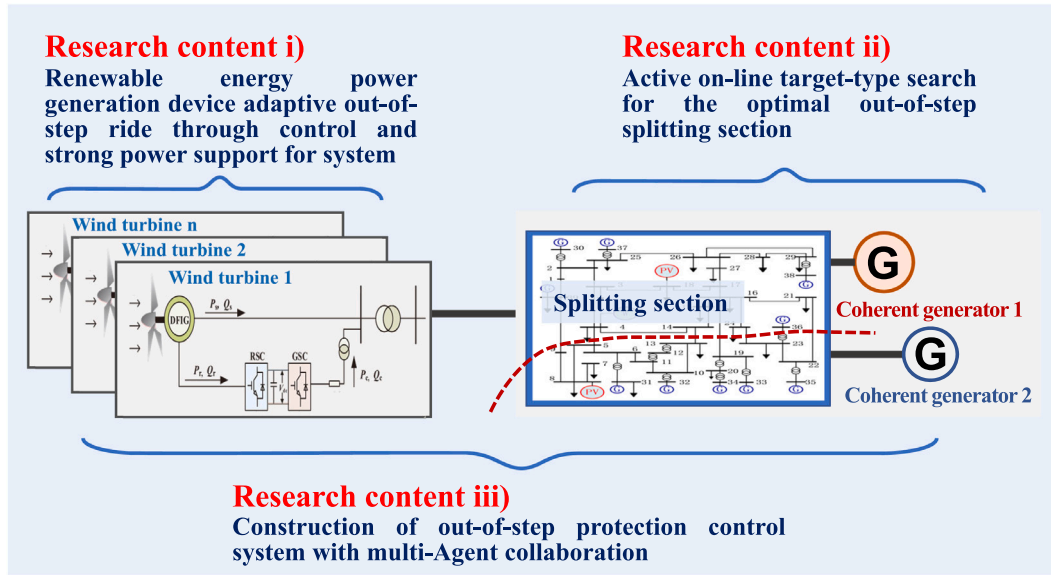


Fig. 16. Interrelationships and difficulties of follow-up research.

control in out-of-step scenario. Under this background, the author thinks that the following three tasks need to be done urgently. This will be discussed in detail below (see Fig. 16).

(i) Renewable energy power generation device adaptive out-of-step ride through control and strong power support for system

Accurate electrical quantity measurement is the prerequisite for the renewable energy power generation device reliable ride through. In the traditional high, low and zero voltage ride through situations, the bus voltage amplitude only changes step-wise or the frequency fluctuates slightly. The power grid synchronization information requires by the renewable energy power generation device can be relatively easy to get. However, the out-of-step oscillation is a special state, in which the bus voltage fluctuates sharply, satisfying

$$u(t) = U(f_o(t), t) \cos(2\pi f_s(t) * t) \quad (6)$$

During the whole out-of-step oscillation process, the voltage amplitude U , oscillation frequency f_o , and cycle frequency f_s are all time variables. Specially, in certain time period, the frequency variation can reach 1 Hz/s. The traditional DFT voltage and current detection algorithm introduces a large error due to the fence effect, while the SRF phase lock is difficult to adapt to the signal amplitude frequency change due to the fixed parameters. As a result, the renewable energy power generation device cannot be reliably connected to the grid. It could cause equipment damage and even large-scale disconnection. Therefore, it is necessary to study a strong robust and adaptable voltage and current detection method, which should be able to obtain the accurate measurement under the out-of-step condition where both the amplitude and frequency change greatly at the same time.

For renewable energy power generation device itself, it should overcome the insufficiency of control strategy under the out-of-step oscillation situation. At present, renewable energy power generation device ride-through control strategies are mostly proposed for short-circuit faults or small disturbances, including: (1) excitation converter control scheme, achieving ride through by controlling the inherent capacity of the converter itself; (2) with the help of terminal voltage support device, suppressing the depth of voltage drop or reducing the drop rate; (3) improving the combination of excitation control and hardware protection, and so on [104–106]. However, under the out-of-step oscillation situation, the amplitude and frequency of the voltage at grid-connected point change greatly, which makes the renewable energy grid-connected model present strong non-linearity. It is difficult to control the rotor over-current and the parallel capacitor pumping

voltage. The related impact has been concerned, but there is no good solution yet. In addition, few renewable energy control structure includes the stability control link, which participates in the system frequency stability control. How to realize the active and reactive support for the unstable system under the out-of-step oscillation situation has not yet been clarified, a breakthrough is needed!

(ii) Active on-line target-type search for the optimal out-of-step splitting section

Fast online searching for the optimal out-of-step splitting section is an extremely important task for the out-of-step splitting control. It is related to the success or failure of the entire splitting process. As discussed above, the existing wide-area information based out-of-step splitting method first classifies the generator sets into coherent groups, and then uses graph theory and other methods to search. They adopt the search strategy from outside to inside, the search scope covers the entire network. They face the serious problem of “dimensional disaster”, which can only run offline when the system becomes complex. Besides, after the large-scale access of renewable energy sources, the instability mode changes rapidly, and easily evolves from single instability mode to complex multi-instability mode, the difficulty of online search increases. The contradiction between the network complexity and splitting rapidity requirements become more obvious.

In order to solve the contradiction between the rapidity and stability of out-of-step splitting as discussed above, the fast task is to solve the problem of accurate identification on the system operation states (Normal, Fault, Out-of-step oscillation, Re-fault in the out-of-step process), because it directly determines the selection of system control strategy. The key of improving its accuracy and speed is the mastery of the mechanism and characteristics of out-of-step oscillation. However, large scale access of renewable energy seriously affects the dynamic behavior of out-of-step power system, such as the migration of the out-of-step center, the evolution of the out-of-step oscillation mode, etc. It easily leads to the error of system operation state judgment and makes the existing out-of-step protection schemes refuse to operate or mal-operate. Therefore, it is suggested to analyze the coupling mechanism between renewable energy power generations and the out-of-step power system, and draw the characteristics of electrical quantity change during the out-of-step oscillation process for the renewable energy power system. Moreover, it should explore new principles to accurately and rapidly judge the out-of-step oscillation mode and locate the out-of-step center for the further splitting operation.

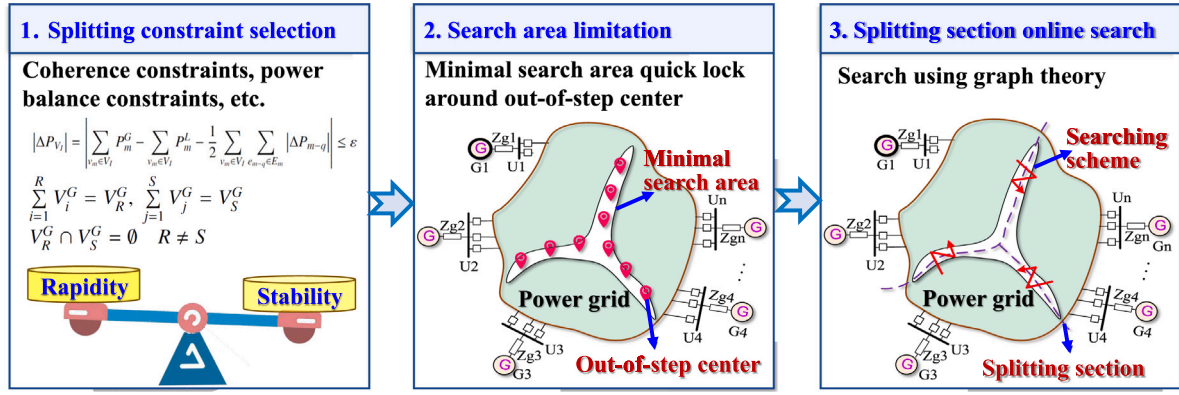


Fig. 17. Active on-line incremental search of out-of-step splitting section.

On the other hand, it should break the constraints of traditional search strategies, and combine the advantages of the first and second class of out-of-step splitting control. Specifically, it could study a “targeting” optimal splitting section search algorithm. Like Fig. 17 shows, it could search find some points with special physical meaning as the initial splitting section, and take it as the center and search from inside to outside. By this way, the search scope is limited and the search speed would increase. The feasibility of this search idea has been discussed in literature [107], and an out-of-step splitting section search method suitable for single instability mode is tentatively given. Different from the traditional wide-area information search strategy shown in Fig. 15, this method makes full use of the natural attribute of the out-of-step center as the frequency demarcation point of the system, makes the electrical components where they are located as the initial splitting section, and constructs a search tree based on this to perform a quick search. This method directly left the link of generator coherence recognition out, and limits the search range to the vicinity of out-of-step centers. The search difficulty and scope does not increase with the increase of the complexity of the power network, which effectively solves the “dimension disaster” problem. This is a new thinking and bold attempt on the problems existing in the existing out-of-step splitting section search methods. But this method adopts the single island search mode, and uses the load shedding and other means to ensure the final stability. This can guarantee the power balance constraints is meet, but it cannot declare it is the optimal scheme. Especially in the case of multiple instability modes, the optimality needs further research. In addition, since the search process is very fast, the splitting constraint can be further increased in order to achieve better splitting effect. All in all, this is a bold attempt. The author hopes that it can be continuously improved in subsequent research, and the third out-of-step solution can be formed with the joint efforts of scholars.

(iii) Construction of out-of-step protection control system with multi-Agent collaboration

As the content of system-level protection and control, the above two aspects are the extremely important but most basic technical links. In fact, completing the out-of-step splitting control should also include other technical contents such as fault unblocking for the distance protection, voltage balance control after splitting, and security domain evaluation, etc. However, the completion of these functions are not independent, it needs the frequent exchange of the wide-area measurement information and control information, the assisted technical links like data acquisition, processing, sharing. Obviously, such complex out-of-step splitting control process can no longer be simply completed by an independent control body. It requires multiple control bodies work together under the unified deployment of the overall control unit. Therefore, it is necessary to construct a wide-area information based online out-of-step splitting control system. Literature [108] gives the concept of Wide Area Monitoring, Protection and Control (WAMPAC). It is a kind of protection system involving all kinds of protections

in power grid, such as the out-of-step protection, the line back up protection and the transformer protection, etc. It is not a protection system specially for the out-of-step oscillation problem and did not study out-of-step detection and splitting section search. Literature [109] introduces the System Integrity Protection Scheme (SIPS), and gives one optimal bus-splitting scheme which could eliminate or reduce power system overloads and prevent out-of-step occurs in system. It is also not a special out-of-step protection system, but one for the preventive control. However, the above two articles have very important reference significance for the out-of-step protection system construction.

In order to develop such type of out-of-step splitting control system, the research can be done from the following aspects: (1) System definition and control architecture selection. The system definitions on the work goals and tasks are the “programme” of the out-of-step splitting control system, the control architecture is the “main body”, and the communication architecture is the “nerve center”. Above three parts are closely related to the overall performance of the entire out-of-step splitting control system. As an effective solution to systemic problems, multi-agent systems have been used in power system protection and control. In such a system, a certain operational task can be completed by a single Agent with mathematical calculations or rule reasoning, and through interacting with other Agents or objects, a systematic level power system problem could be completed. However, there is no mature plan for out-of-step splitting control until now. To construct such a system, the tasks of out-of-step splitting control, time limit, cooperation mechanism with other system protection and response priority, communication architecture, etc. should be clarified according to the requirements and actual needs of out-of-step splitting control. (2) Multi-source heterogeneous data fusion and reduction. The out-of-step control system needs to comprehensively utilize large amount of information, involving different types of electrical components, electrical quantities, change times, network connection methods, network status information, etc. For a specific control unit, only a specific form combination of data is required. In order to improve the system operational efficiency, the above-mentioned multi-source heterogeneous data needs to be processed to meet applicable requirements for different scenarios. (3) System-level integrated prevention and control. The out-of-step splitting control system should not only include the technical content of suggestions (i) and (ii), it also needs to incorporate dynamic security domain technology to comprehensively judge the safety and stability margin of the split islands, and carry out preventive control on serious accidents to prevent secondary islanding. On the other hand, it is necessary to comprehensively utilize system-level control technologies such as multi-machine output coordination and multi-stage load shedding to achieve rapid power recovery and safe and stable control of local isolated grids.

(iv) Application of AI technology in out-of-step splitting control

Until now, AI technology has been gradually applied in out-of-step splitting control. In literature [110], an innovative out-of-step

protection algorithm using wavelet transform and deep learning is presented. A two-layer feed-forward network with the new method is trained with 27,008 continuous samples of 48 different features. It can give an automatic, setting free classification for short circuit, synchronous oscillation, and out-of-step oscillation within a half cycle. Literature [111,112] propose two out-of-step oscillation prediction methods based on the deep Q network (DQN) disturbance observer. They utilize the estimation information to predict whether a short circuit would result in out-of-step oscillation and execute splitting operation. In literature [113], an intelligent trajectory recognition based out-of-step protection method with support vector machine (SVM) is presented. In literature [114], a PMU-based polygon-shaped graphical algorithm based out-of-step protection is proposed to detect complex out-of-step oscillation. Besides, decision tree classifiers [115], artificial neural network (ANN) [116], deep convolutional neural network (DCNN) [117], and Bayesian technique [118], are utilized to predict out-of-step oscillation events. At present, the application of AI technology is mostly focused on the identification process of out-of-step oscillation. In facts, AI technology would own more applications in the future. Here's a little look into the future. For example, according to the conclusion of literature [119], the frequency of the system changes drastically near the out-of-step splitting section, while on the contrary, the voltage and current frequency of the bus far away from the out-of-step splitting section change very little. Therefore, using the bus voltage and current frequency actually sampled by the PMU, a frequency "cloud map" of the entire network can be quickly formed. Then, by using image recognition technology and AI technology, combined with the traditional out-of-step splitting control analysis conclusions, the rapid determination of the out-of-step splitting section can be achieved. For another example, in terms of predicting out-of-step oscillation, deterministic learning theory methods can be used for dynamic modeling, and intelligent learning methods can be used to prejudge the occurrence of out-of-step oscillations, providing valuable time for splitting control. All in all, the application of AI technology will make the out-of-step splitting control scheme more complete and the performance even better.

5. Conclusions

This review is written under the background that out-of-step splitting control has achieved many achievements after years of development. It is used to analyze the technical routes of existing methods, summarize their advantages and disadvantages, and give some suggestions. It is hoped to provide some guidance for the subsequent research work.

First of all, for the problem of out-of-step splitting control of traditional complex power grids, it points out that the core problem of existing methods is the contradiction between rapidity and optimality of splitting under multiple network nodes and multiple stability constraints. For example, the traditional out-of-step splitting method based on local measurement information can quickly perform splitting after the system is out-of-step, but it does not consider the constraints such as power balance. The method based on wide-area information fully considers the stability of splitting islands, but it is difficult to apply online due to many constraints and large amount of calculation. Through analysis, this review finds although neither of the above two types of methods is the optimal solution, the advantages and disadvantages are complementary. Through the integration of technical routes, a new type of out-of-step splitting control strategy is expected to form, which can effectively solve the contradiction between rapidity and optimality of splitting.

In view of the reality of large-scale renewable energy integration into the power grid, this review points out the challenges that it will bring to the out-of-step splitting control through literature and theoretical analysis, which can be attributed to the following two problems: How to realize renewable energy protection and power

support under out-of-step situation? How to deal with the severe requirements of large-scale access of renewable energy on reliability, rapidity and applicability of out-of-step splitting control? Based on this, some corresponding suggestions are proposed.

CRedit authorship contribution statement

Shuai Zhang: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

No conflict of interest exists in the submission of this manuscript. I would like to declare that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part.

Data availability

No data was used for the research described in the article.

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